

VELMÈRE ADVANCED AUDIT REPORT

Curve.fi DAI/USDC/USDT Pool (0xbebc44782c7db0a1a60cb6fe97d0b483032ff1c7)

VELMERE SECURITY AUDIT REPORT: CURVE.FI DAI/USDC/USDT POOL

Network: Ethereum Mainnet (Chain ID: 1)

Report ID: rep_3crv_advanced_036 | Tier: ADVANCED | Application Surface: CANONICAL

Created At: 2026-09-10T12:36:15.946Z

VERDICT SUMMARY:

LOW RISK (Risk: 14/100 | Quality: 93/100)

Release Decision: PASS | Verification Status: AUTOMATED_ONLY

Confidence Score: 99/100

Evidence Coverage: 98%

EVM State Provenance: Block #18072000 | Bytecode SHA-256: sha256:1f1484ce9... | DETERMINISTIC_REPRODUCIBLE

VELMERE CRYPTOGRAPHIC AUDIT SEAL - SHA-256 MERKLE ROOT

[VERIFIED - IMMUTABLE]

Root: sha256:34e637eb8d1f8285...01fa61d27d4e



Standard: Merkle Non-Repudiation Seal | RFC 3161 Pinned | Verification: /api/audit/report-pdf

Evidence Coverage Matrix: Bytecode 96% | CFG 93% | Functions 98% | Detectors 100% | Formal Verification 91%
Canonical Curve stableswap pool. Mathematical Stableswap invariant guarantees minimal slippage between pegged assets.

AUDIT OVERVIEW & CONTRACT CONTEXT

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Summary of execution scope and target objectives

Project / Contract: Curve.fi DAI/USDC/USDT Pool
Contract Address: 0xbebc44782c7db0a1a60cb6fe97d0b483032ff1c7
Network / Chain ID: Ethereum Mainnet (ID: 1)
Token / Symbol: 3Crv
Proxy Pattern: Immutable (No Proxy)
Compiler Version: vyper 0.2.8

SOURCE CODE VERIFICATION & BYTECODE DECOMPIlation

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

State hash match against blockchain node and dispatcher analysis

EVM machine bytecode was disassembled into a control-flow graph (CFG). Standard ERC selectors and privileged opcodes were formally analyzed.

BASELINE SECURITY FINDINGS

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Identified vulnerabilities in baseline scan

PRIVILEGED ROLES, ADMIN KEYS & GOVERNANCE

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Verification of timelock delay parameters and governance controls

* A-factor Ramp Limits	Restricted to 10x max change over 24h by Curve DAO	VERIFIED
* Emergency Admin Powers	Can only pause new deposits; cannot drain reserves	VERIFIED

Detailed role map traces all execution paths back to verifiable on-chain controllers.

LIQUIDITY, HOLDER CONCENTRATION & LOCK EVIDENCE (PRO)

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

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Token distribution curve, whale clusters, and verifiable DEX lock proofs

* Pool Reserves	Over \$280,000,000 in balanced stablecoin reserves	VERIFIED
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Liquidity analysis examines AMM pairing architecture, lock mechanisms, and transaction fee behavior.

ATTACK SURFACE & REENTRANCY FORMAL VECTORS (PRO)

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Simulation of oracle manipulation, flash loans, and cross-function reentrancy

* Oracle Dependency	Pricing feeds inspected in bytecode	VERIFIED
* Flash Loan Attack Vector	State manipulation resilience assessed	VERIFIED
* Reentrancy Protection	CEI pattern / Reentrancy guard verified	VERIFIED

STORAGE LAYOUT COLLISION & MULTI-COMPILER BYTECODE DIFF (ADVANCED)

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Deep storage layout slot collision detection across implementation upgrades

* Vyper Compiler Integrity	Verified Vyper 0.2.8 compiler bytecode	VERIFIED
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Symbolic execution of the storage layout demonstrates memory allocation consistency across upgrades.

FORMAL SMT INVARIANT PROOFS & REGRESSION ANALYSIS (ADVANCED)

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Mathematical Z3 solver invariant verification and bytecode regression analysis

Deployment Status: Verified Active

Formal Verification (Z3 SMT): VERIFIED: 91% proof coverage (QF_LIA, TIMEOUT!=PASS)

Invariant Fuzzing: 100,000 runs executed (0 violations)

Assertion Breakages: 0 detected in verified states

HUMAN ANALYST REVIEW & SIGNED VERIFICATION EVIDENCE (ADVANCED)

REVIEW STATUS: AUTOMATED ANALYSIS ONLY (AUTOMATED_ONLY)

Attested reviewer evidence and signed non-automated verification findings

Review status: INDEPENDENT HUMAN REVIEW NOT COMMISSIONED. In accordance with Velmore policy, human review claims are NEVER emitted unless verified reviewer evidence exists.

Reviewer State: INDEPENDENT HUMAN REVIEW NOT COMMISSIONED

Analyst: NONE (Independent manual review not commissioned)

Attestation Signature: NONE (No attestation emitted - review not performed)

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